Course: BM20A4703 Partial Differential Equations with Applications.
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Worksheet 4

Exercise 4.1

(i) [5] Using definition, find a weak derivative of the function $u(x) = |x|$ for $x \in \mathbb{R}$.
(ii) [10] Prove that the following function does not have a weak derivative

$$u(x) = \begin{cases} x & \text{if } 0 < x \leq 1 \\ 2 & \text{if } 1 < x < 2. \end{cases}$$

Hint: Show that there is no function $v \in L^1_{\text{loc}}(0,2)$ such that $\int_0^2 u\varphi' \mathrm{d}x = -\int_0^2 v\varphi \mathrm{d}x$, by contradiction.

Exercise 4.2

For $a \in \mathbb{R}$, define the function $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \frac{1}{|x|^a}$$

(i) [10] Suppose that $\epsilon > 0$ and let $\varphi^\epsilon \in C_c^\infty(\mathbb{R}^n)$ be a cut-off function that is equal to one in $B_\epsilon(0)$ and zero outside $B_{2\epsilon}(0)$ (see Notations page for the definition of open ball $B_r(x)$). Let

$$f^\epsilon(x) = \frac{1 - \varphi^\epsilon(x)}{|x|^a}.$$

Show that for every $i = 1, \ldots, n$, there exists a constant $C > 0$ independent of ϵ such that

$$|\partial_{x_i} f^\epsilon(x)| \leq \frac{C}{|x|^{a+1}}.$$

(ii) [10] Use (i) to show that f is weakly differentiable if $a + 1 < n$ with weak derivative

$$\partial_{x_i} f(x) = -\frac{a}{|x|^{a+1}} \frac{x_i}{|x|}.$$

Hint: Dominated Convergence Theorem may help.

Exercise 4.3

Let $U = B_1(0)$ be the unit ball in $\mathbb{R}^2$. Let $u : U \to \mathbb{R}$ defined by

$$u(x) = \log\log\left(1 + \frac{1}{|x|}\right).$$

(i) [10] Is u a bounded function in U? Prove that $u \in L^2(U)$.
(ii) [10] Prove that $u \in H^1(U)$.

Hint: You may use polar coordinate change of variable.

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